DRDO Internship Assignment Submission

Project Title: Vulnerability Assessment of DVWA Web Application Using Kali Linux and MCP Server Integration with AI Client (Claude)

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1. Introduction

This project demonstrates the process of conducting a vulnerability assessment on the DVWA (Damn Vulnerable Web Application) hosted on Kali Linux using penetration testing tools integrated through an MCP server. The testing environment is connected to an AI client (Claude Desktop), enabling automated vulnerability analysis and report generation.

The objective is to identify security loopholes, particularly focusing on SQL Injection and other web vulnerabilities, and generate a structured vulnerability report.

2. Tools & Environment Setup

Tools Used

- VirtualBox
- Kali Linux (Guest VM)
- DVWA Web Application
- Python MCP Server (API tools automation)
- Claude Desktop (AI Client)
- Terminal & Curl

System Configuration

- Host OS: Windows 10
- Guest OS: Kali Linux on VirtualBox
- Network Mode: Bridged Adapter
- MCP Server IP: 192.168.1.8
- Server Port: 5000

3. Steps Performed

Step 1 — Install Kali Linux in VirtualBox

Kali Linux ISO was downloaded and installed on VirtualBox. Base configuration and networking setup were completed.

Step 2 — Installing MCP Server on Kali

The MCP server repository was cloned and configured with required Python dependencies.

```
git clone https://github.com/Wh0am123/MCP-Kali-Server.git
cd MCP-Kali-Server
python3 -m venv venv
source venv/bin/activate
pip install -r requirements.txt
```

The server was started using:

```
python3 kali_server.py --ip 0.0.0.0 --port 5000
```

Step 3 — Enabling Network Communication Between Windows Host & Kali VM

VirtualBox network adapter configured to **Bridged Adapter**, resulting IP obtained using:

```
ip a
```

Example allocated IP:

```
inet 192.168.1.8/24
```

Tested MCP server endpoint:

```
curl http://192.168.1.8:5000/api/tools
```

Step 4 — Configured AI Client (Claude Desktop)

Claude Desktop configuration file updated:

```
{
  "mcpServers": {
    "kali_mcp": {
      "command": "python3",
      "args": [
        "/home/kali/MCP-Kali-Server/kali_server.py",
        "--ip", "192.168.1.8",
        "--port", "5000"
      ]
    }
  }
}
```

Claude connected successfully to the MCP server.

Step 5 — Installation of DVWA

DVWA was cloned and hosted locally:

git clone https://github.com/digininja/DVWA.git

DVWA database setup and configuration completed, app accessible at:

http://localhost/dvwa

4. Vulnerability Assessment With AI Assistance

After configuring DVWA security level to **Medium**, the MCP-connected AI client was instructed to:

- Scan DVWA
- Identify vulnerabilities
- Highlight exploit possibilities

Key Vulnerabilities Identified

Vulnerability	Description	Status
SQL Injection	Improper sanitization of input fields	Found
Command Injection	Web UI accepts unsanitized commands	Found
XSS	Reflected cross-site scripting	Found
Weak Authentication	Default login credentials	Found

5. Recommendations & Mitigation

Risk	Recommended Fix
SQLi	Use prepared statements, parameterized queries
XSS	Apply output encoding, use CSP policy
Command Injection	Validate & whitelist command inputs
Weak Auth	Enforce strong passwords & MFA

6. Conclusion

This assessment successfully established a connected environment between Kali Linux and Claude using MCP, identified multiple high-risk vulnerabilities in DVWA, and generated automated and human-assisted vulnerability insights.

The project demonstrates capability in:

- Network configuration & automation tool integration
- Web security assessment fundamentals
- Practical penetration testing workflow

Declaration

I hereby declare that this project was completed by me as a requirement for the DRDO internship vulnerability assessment submission.

Signature: *Alisha Karwal*

Date: *28 Nov 2025*